



Pranaya Pratik Das

Physics, Research Scholar

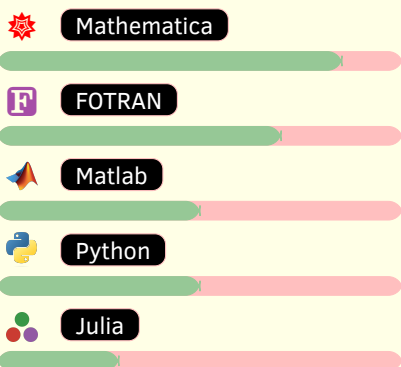
December 22, 1992
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 pranaya_phy@outlook.com

Online Platforms

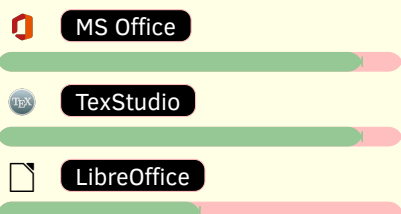
Pranaya Pratik Das
 Pranaya-Das
 0000-0002-6025-7719

Skills

> </> Programming Software



> ✎ Text editing Software



> 🖥️ Operating Systems



Education

Academic Curriculum

- 🌟 10th. 📅 June 2008 BSE, Odisha
Achievement: 100% in Mathematics
- 🌟 +2 Science 📅 June 2010 CHSE, Odisha
- 🌟 B.Sc. in Physics 📅 July 2010 – June 2013 BJB Auto. College, Utkal University
- 🌟 M.Sc. in Physics 📅 August 2014 – August 2016 CBSH, OUAT
- 🌟 Ph.D. in Physics 📅 July 2019 – December 2025 NIT Rourkela
- Thesis Title: *Diagnosis of quantum chaos in perturbed quantum wells and billiards*

Awards and Achievements

Scholarships:

- 🌟 P.G. Meritorious Scholarship (2014-16), Institute of Mathematics and Applications (IMA).
- 🌟 Medhabruti Scholarship (2014)

Certificates:

- 🌟 IAPT(2012)
- 🌟 Spring College in the Physics of Complex Systems. Awarded by ICTP.
- 🌟 Secrets of getting published in high impact factor journals. Awarded by Wiley.

Qualified Entrances:

- 🌟 TIFR (2015-16)
- 🌟 GATE (2019-21)

Research & Publications

Recent Publications

- 2025: Pranaya Pratik Das and Biplab Ganguli. “Harmonic oscillator under nonlinear spatial perturbations: A study via Out-of-Time-Order Correlators”. (On going)
- 2025: Vinesh Vijayan, Pranaya Pratik Das, K Hariprasad and P Sathish Kumar. “Cyclically Symmetric Thomas Oscillators As Swarmalators: A model for Active Fluids and Pattern Formation.”. **Communications in Nonlinear Science and Numerical Simulation Volume 152, Part B, January 2026, 109216**
DOI: <https://doi.org/10.1016/j.cnsns.2025.109216>
- 2025: Pranaya Pratik Das and Biplab Ganguli. “Out-of-Time-Order Correlation in perturbed quantum wells”.
Eur. Phys. J. D (2025) 79 :74
DOI: <https://doi.org/10.1140/epjd/s10053-025-01025-7>
- 2025: Pranaya Pratik Das, Tanmayee Patra, and Biplab Ganguli. “Manifestations of chaos in billiards: the role of mixed curvature”.
DOI: <https://doi.org/10.48550/arXiv.2501.08839>
- 2024: Bhaskar Shukla, Pranaya Pratik Das, David Dudal, and Subhash Mahapatra. “Interplay between the Lyapunov exponents and phase transitions of charged AdS black holes.”.
Phys. Rev. D 110, 024068
DOI: [10.1103/PhysRevD.110.024068](https://doi.org/10.1103/PhysRevD.110.024068)
- 2022: Vinesh Vijayan, & Pranaya Pratik Das. “Dynamics of a charged Thomas oscillator in an external magnetic field”.
Physica Scripta, 97(11), 115207.
DOI [10.1088/1402-4896/ac99ab](https://doi.org/10.1088/1402-4896/ac99ab)

Research Interest

- 🌟 Classical Chaos 🌟 Quantum Scars 🌟 Billiards Dynamics
- 🌟 Quantum Chaos 🌟 Quantum Speed Limits 🌟 Black Hole Phase Transition

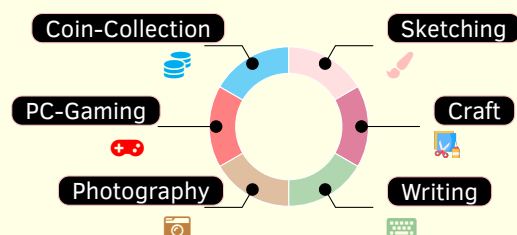
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About Me

I am a research scholar specialising in quantum chaos and nonlinear dynamics, focusing particularly on out-of-time-order correlation (OTOC), spectral complexity, scrambling diagnostics, and quantum scars. By collaborating on projects related to black hole phase transitions, chaos near black holes, and the swarming dynamics of active particles, my work bridges multiple domains.

Hobbies



Languages



References

- Prof. Biplab Ganguli**
Department of Physics and Astronomy,
NIT Rourkela
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biplab62g@gmail.com
- Prof. Subhash C. Mahapatra**
Department of Physics and Astronomy,
NIT Rourkela
mahapatrasub@nitrkl.ac.in
subhashmahapatra@gmail.com
- Prof. Mithun Biswas**
Department of Physics and Astronomy,
NIT Rourkela
biswasm@nitrkl.ac.in

Teaching Assistance

Laboratory

- TA in Thermal Physics Lab 2022–2023, Spring Semester** Prof. B. Ganguli
 - Conducted various classes, experiments, evaluated copies and taken vivas.
- TA in B.Tech Physics Lab 2022–2023, Spring Semester** Prof. B. Ganguli
 - Conducted various classes, experiments, evaluated copies and taken vivas.
- TA in B.Tech Physics Lab 2022–2023, Autumn Semester** Prof. I. Banarjee
 - Conducted various classes, experiments, evaluated copies and taken vivas.
- TA in B.Tech Physics Lab (Online) 2021–2022, Spring Semester** Prof. S. K. Bisoi
 - Conducted various online classes, seminars and evaluated copies.
- TA in B.Tech Physics Lab (Online) 2021–2022, Autumn Semester** Prof. P. Mahanandia
 - Conducted various online classes and evaluated copies.

Theoretical

- Co-supervised a M.Sc. project 2023–2024** Prof. B. Ganguli
 - Successfully co-supervised an M.Sc. project for Zubair Ahmad Kumar (422PH2069).
 - Successfully co-supervised an M.Sc. project for Vivek Sheoran (422PH2082).
- Co-supervised a M.Sc. project 2022–2023** Prof. B. Ganguli
 - Successfully co-supervised an M.Sc. project for Karishma Kujur (421PH2125).
 - Successfully co-supervised an M.Sc. project for Ayush Sahu (418PH5033).

Conferences and Schools

- 60 Years of DFT: Advancements in Theory & Computation 2024** IIT Mandi, India
 - Poster Presentation
- HPC Symposium 2024** NIT Rourkela, India
 - Poster Presentation
- Integrability, Deformations and Chaos 2023** OIST, Onna, Okinawa, Japan
 - Attended online
- School on Quantum Chaos 2023** ICTP-SAIFR, São Paulo, Brazil
 - Attended online
- Complex Lagrangian Problems of Particles in Flows 2022** ICTS-TIFT, India
 - Attended online
- Spring College in the Physics of Complex Systems 2022** ICTP, Trieste, Italy
 - Attended online